

9 - The Periodic Table

19/12/24

Periodicity of physical properties of the elements in Period 3

⇒ atomic radius

- decreases gradually, because nuclear charge increases, thus outer electrons are more attracted towards the nucleus, making the atom smaller

⇒ ionic radius

- cations (Na, Mg, Al, Si) are smaller than their respective atoms because by giving electrons, they lose a shell. The remaining electrons are more strongly attracted to the same nuclear charge
- anions (P, S, Cl) are bigger than their respective atoms because they have more electrons than protons so less attraction to the nucleus. The added electrons also create a repulsion which expands the ion

[anions are bigger than cations because they have one more shell]

⇒ melting point

- increases from Na → Al because the strength of the metallic bond increases
- Si has the highest MP because it has a giant covalent structure, requiring high energy to overcome
- P, S, Cl, Ar have a lower MP because their simple molecular structures have weak van der Waals forces
S < P < Cl < Ar because these elements exist as S₈, P₄, Cl₂, Ar. S₈ has the most electrons thus stronger van der Waals forces

⇒ electrical conductivity

- Na → Al are metals, Si is a semi-metal & P → Ar are non-metals
- elec. conductivity is highest in metals, lower in semi-metals & zero in non-metals (most are insulators)
- depends on number of electrons contributed by per atom to the sea of delocalised electrons. Increases from Na → Al, Si is a semi-conductor, P → Ar do not have free mobile electrons to carry the electricity

Reactions with oxygen



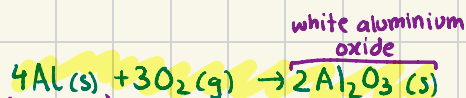
→ burns with an orange-yellow flame

white sodium oxide



→ burns with a white flame

white magnesium oxide

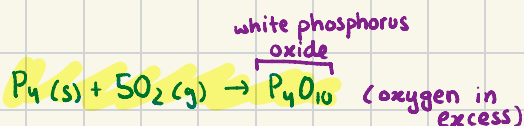


powdered aluminium

white aluminium oxide

→ burns with a white flame

→ when aluminium is exposed to air, it forms an oxide layer to prevent the aluminium from reacting

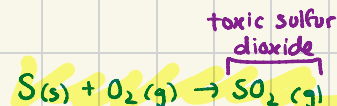


→ burns with a white/yellow flame

→ clouds of white phosphorus oxide form

white phosphorus oxide

(oxygen in excess)



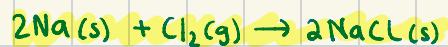
→ burns with a blue flame

→ further oxidation would give sulfur trioxide

toxic sulfur dioxide

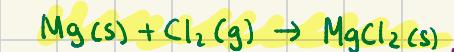
* silicon has a slow reaction & Chlorine and Argon don't react with oxygen

Reactions with chlorine



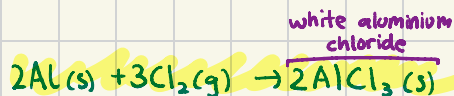
→ Cl₂ is a green-yellow gas
→ burns with an orange-yellow flame

white sodium chloride



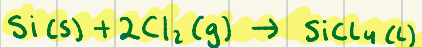
→ burns with a white flame

white magnesium chloride



→ burns on heating to form ionic aluminium chloride

white aluminium chloride



→ burns with a white flame

colorless silicon(IV) chloride



(chlorine in excess)

yellow phosphorus(V) chloride

∴ when chlorine is limited, PCl₃ forms

* Sulfur burns in chlorine gas to form disulfur dichloride but we don't need to know that
* Argon does not form a chloride

20/12/24

	Na ₂ O	MgO	Al ₂ O ₃	P ₄ O ₁₀	SO ₂ /SO ₃
Bonding	ionic	ionic	ionic/covalent (amphoteric)	covalent	covalent
Structure	giant ionic	giant ionic	giant ionic	simple molecular	simple molecular
ox. number of period 3 element	+1	+2	+3	+5	+4/+6

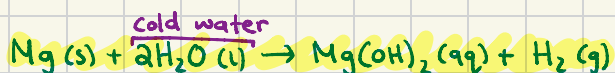
	NaCl	MgCl ₂	AlCl ₃	SiCl ₄	PCl ₃ /PCl ₅
Bonding	ionic	ionic	covalent	covalent	covalent
Structure	giant ionic	giant ionic	simple molecular	simple molecular	simple molecular
ox. number of period 3 element	+1	+2	+3	+4	+3/+5

Reactions with water



→ sodium catches fire in cold water

→ violent exothermic reaction



→ reacts slowly, making H₂ & a weakly alkaline magnesium hydroxide solution



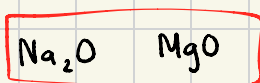
→ reacts rapidly

Flame colors

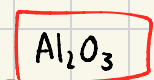
Na yellow flame
Mg white flame
Al white flame

Si ———
P yellow-white flame
S blue flame

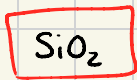
⇒ bases are usually metal oxides/hydroxides



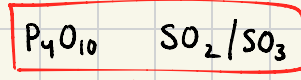
↓
basic
oxides
(ionic oxides)



↓
amphoteric
oxide

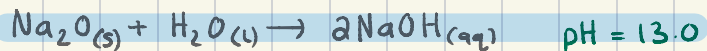


↓
weakly
acidic
(inert)



↓
acidic
oxides
(covalent oxides)

Reaction with water



→ exothermic reaction producing strongly alkaline solution of NaOH



→ slight reaction (almost insoluble) producing weakly alkaline solution of $\text{Mg}(\text{OH})_2$

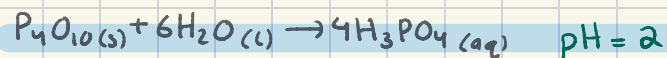
$\text{Al}_2\text{O}_3 \rightarrow$ insoluble in water

* can react with acid & base

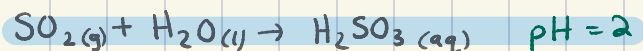
→ doesn't react/dissolve in water due to its high lattice energy

$\text{SiO}_2 \rightarrow$ insoluble in water

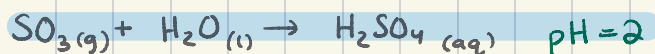
→ doesn't react/dissolve in water due to its strong covalent bonds



→ forms phosphoric (v) acid

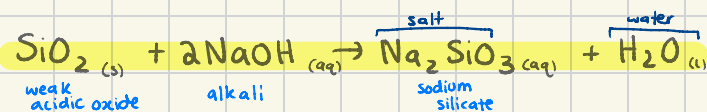
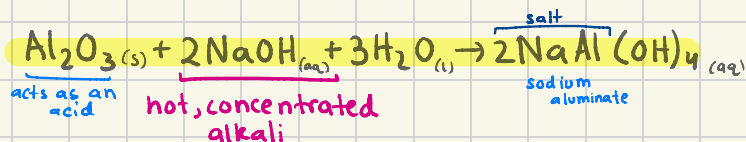
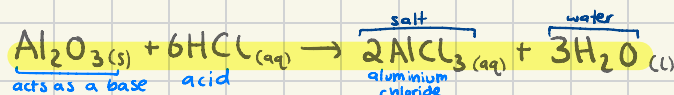
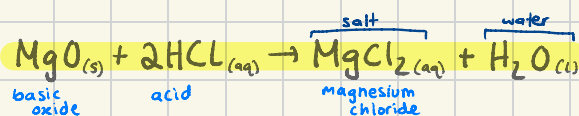
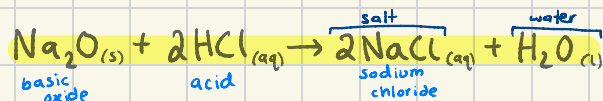


→ forms sulfurous acid

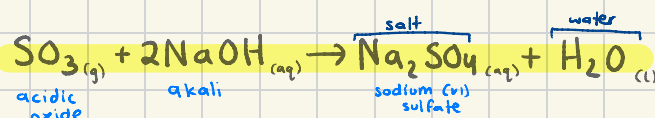
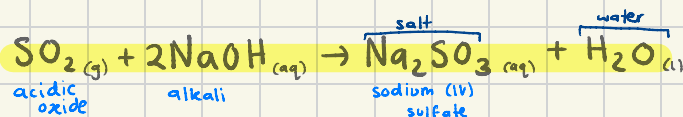
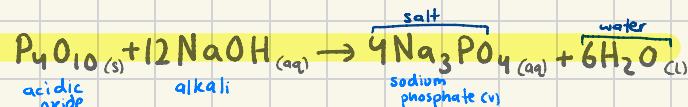


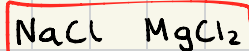
→ violent reaction forming sulfuric acid

Acid/Base behaviour



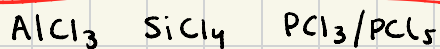
only reacts under extreme conditions (high temp./pressure)





↓
ionic
chlorides

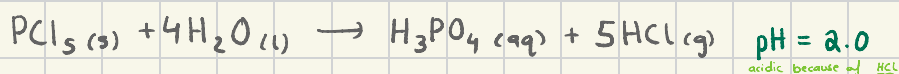
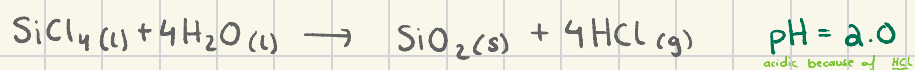
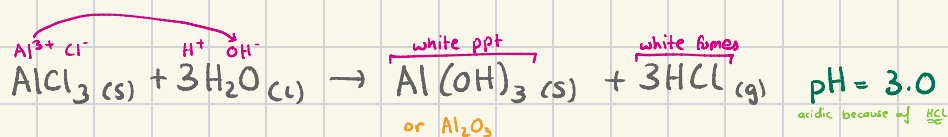
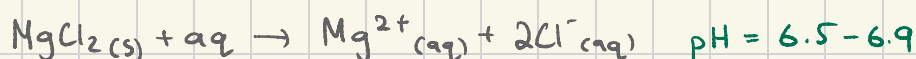
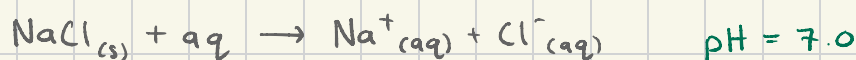
dissociate in water
(no reaction)



↓
covalent
chlorides

react with water
(hydrolysis reaction)

Hydrolysis reaction
water molecule breaks down



observation

: white solid dissolves to form a colorless solution

observation

: chlorides react with water, giving off white fumes of HCl

Q] Element X forms a chloride, which reacts with water to form a solution of pH 1. It forms an oxide which has the melting point of 1610°C. The oxide doesn't dissolve in or react with aqueous sodium hydroxide. Deduce the position of element X in The Periodic Table.

→ its chloride reacts with water to form acidic solution = not group 1/2, has to be group 3-7

→ high M.P of oxide = giant structure & not from period 2 (e.g CO₂ is a gas)

→ doesn't dissolve in NaOH = not Al₂O₃ ∴ giant covalent structure

Element X is in group 3-7 and period 3 or lower

Q] Selenium is in group 16 & period 4 of the Periodic Table. Predict some physical and chemical properties of Se.

→ group 16 = non-metal with a simple molecular structure

Physical ⇒ simple molecular structure: low melting point, doesn't conduct electricity & is insoluble in water

Chemical ⇒ reacts with chlorine to form SeCl₄, reacts with oxygen to form SeO₂

deduced by comparison to sulfur, an element in the same group

*elements in the same group have similar properties